

Earth Rover Article #3 Context Brief - STM32 MCB Integration

Project Narrative Status

Previous Articles:

1. **Article #1:** System architecture overview - multi-MCU rover concept, ESP-NOW wireless, STM32 motor control, SPI communication plan
2. **Article #2:** SPI implementation fundamentals - ESP32-C3 SPI learning, ISR design, FreeRTOS queues, logic analyzer debugging, successful TX→RX communication with test data ("LIGHT_ON")

Current Development Phase: Integration of STM32L432KC Motor Control Board (MCB) into the established TX→RX communication chain, creating full TX→RX→MCB data flow.

Technical Context for Article #3

Starting Point:

- ESP32-C3 TX/RX pair communicating reliably via ESP-NOW
- SPI communication working between RX and test scenarios
- Logic analyzer and ESP_LOG output synchronized and trusted
- ISR and FreeRTOS queue architecture established

Integration Challenge:

- Add STM32L432KC as SPI peripheral to ESP32-C3 RX (controller)
- Implement full-duplex SPI communication
- Handle real-time motor control requirements
- Navigate DMA implementation for high-speed transfers
- Resolve timing and bit ordering complexities

Key Technical Elements to Cover:

STM32 Development Environment

- Transition from ESP-IDF ecosystem to STM32 development
- STM32CubeMX usage for initial code generation
- CMake integration vs. STM32CubeIDE workflow decisions
- OpenOCD and GDB debugging return (contrast with ESP-IDF approach)

SPI Peripheral Implementation

- STM32 as full-duplex SPI peripheral configuration
- Interrupt handling on STM32 side
- Data reception and motor command processing
- Bidirectional communication (commands in, status out)

DMA Integration Challenges

- Why DMA became necessary (timing requirements)
- Buffer management complexity
- Circular buffer implementation
- Interrupt coordination between SPI and DMA
- Buffer overflow mitigation strategies

Timing and Synchronization Issues

- ESP32-C3 vs STM32L432KC timing characteristics
- Clock domain differences
- Setup/hold time violations
- Logic analyzer debugging of timing relationships

Bit Ordering and Endianness

- Data format inconsistencies between platforms
- Byte order resolution
- Protocol definition for reliable data exchange
- Packet structure development

Motor Control Integration

- TB6612FNG driver integration with SPI commands
- Real-time control loop requirements
- PWM generation coordination with SPI communication
- System response time validation

Engineering Methodology Continuation

- Component isolation testing approach

- Logic analyzer validation at each integration step
- Systematic debugging when issues arise
- Documentation discipline maintained in engineering journal

Current Status Leading Into Article

- Basic STM32 SPI peripheral functional
- DMA implementation working but complex
- Timing issues identified and being resolved
- Full TX→RX→MCB data flow established
- Motor control responding to SPI commands

Article Tone and Style

- Continue engineering discovery narrative
- Focus on real integration challenges, not tutorial examples
- Emphasize systematic problem-solving approach
- Show progression from working components to integrated system
- Highlight where theory meets hardware reality

Connection to Broader Portfolio

- Builds on SPI foundation from Article #2
- Demonstrates multi-platform embedded development
- Shows real-time system integration complexity
- Sets up for analog input expansion (potentiometer work)
- Precedes CD4053BE multiplexer discovery that catalyzes DINO CPU interest

Future Article Setup

Article should conclude by establishing:

- Reliable STM32 integration foundation
- DMA-based communication working
- Ready for transition to real analog inputs (potentiometers)
- Sets stage for ADC limitation discovery and multiplexer work

Key Technical Insights to Convey

- Multi-platform embedded development complexity
- Why DMA becomes necessary in real-time systems
- Importance of systematic debugging across hardware platforms
- How component integration reveals new categories of problems
- Value of logic analyzer validation in complex timing scenarios

Note: This phase represents the transition from communication protocol development to real-time system integration - a significant step up in complexity that demands different engineering approaches.